

Biochemistry paper 2 - Nucleic acid notes

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1 Synthesis of Nucleosides

Three strategies: 1. Formation of glycosylic bond by joining sugar onto preformed base. Uses substitution of a nucleophile, joining at C1 of aldose derivative; 2. building heterocyclic base onto preformed nitrogen or carbon substituent; 3. formation of purine base onto an imidazole ribonucleoside.

1.1 Formation of glycosylic bond strategies

1. Heavy metal salts to catalyze the SN reaction onto C1 of a protected sugar (mostly β -anomers and acyloxonium ion. Disadvantages: insolubility of certain heavy metals (e.g., mercury) + instability of halogeno-sugar derivatives. Solution: fusion method using Lewis acids (e.g., TiCl_4 or SnCl_4) where melted 1-acetoxy sugar and suitable base = acceptable nucleosides.

Hilbert-Johnson reaction: alkylation with 2-alkoxypyridine and an halogeno-sugar. Initial product = quaternary salt. Higher temperatures yields condensation intermediate after alkyl halide elimination. Improvement using silyl bases.

Vorbrüggen procedure: 1-acetoxy ribose / 1-chloro sugar $\rightarrow$ electrophile ready to be activated; add Lewis acid (TMSOTf , SnCl_4 or $\text{BF}_3 - \text{Et}_2\text{O}$) to polarize the C1-LG bond, C1 activated; use a silylated nucleobase; nucleophilic attack on C1; desilylation.

2 Acyclonucleotide Synthesis

Acyclonucleoside: nucleoside without sugar ring or with open sugar ring and N-glycosylic bond.

Retrosynthesis of acyclonucleosides: start with heterocycle, attach heterocyclic chain with a leaving group, N-alkylation with base nucleophile and electrophile chain.

Mechanism: Base activation - deprotonation of N9 (purines) or N1 (pyrimidines) leads to nucleophilic nitrogen. Chain activation - converts terminal alcohol into a leaving group, SN2 attack - base backside attack.

3 Phosphate Esters

Orthophosphoric acid $(\text{HO})_3\text{P}=\text{O}$, tetrahedral geometry. Phospho triesters = alkyl or aryl group substituents.

Phosphate hydrolysis pathways: Associative (A):

- Nucleophile attacks phosphorus = pentacoordinate intermediate
- 5-coordinate stabilized by hypervalent bonding
- backside **SN2** attack and leaving group departure

Associative dominates at negatively charged phosphate esters, poor leaving groups, aqueous solution with strong solvation, strong enzymatic stabilization.

Dissociative pathway (D)

- leaving group leaves first = metaphosphate intermediate
- phosphorus = p(V)-like reduced coordination
- nucleophile attacks metaphosphate

4 Dialkyl Phosphate Biosynthesis

Condensation of ATP with monoalkyl phosphate. Cyclic ADP ribose = product of cyclization of NAD^+ with ring closure of C1 of second ribose onto N1 of adenine. Acts as second messenger and calcium signaling in cells.

5 Synthesis of Nucleoside Polyphosphate Esters

Poulter reaction: method for synthesis of nucleoside polyphosphate esters (ADP, ATP, NAD^+) by reacting: nucleoside phosphate ester with phosphorimidazolide or phosphoromorpholidate-activated phosphate, forming P–O–P bonds.

Step 1: $\text{PO}_3^{2-} + \text{carbodiimide} + \text{imidazole} \longrightarrow \text{activated } \text{PO}_3^{2-}$, $\text{PO}_3^{2-} + \text{DCC} + \text{morpholine} \longrightarrow \text{phosphoromorpholidate}$ converting leaving groups. Step 2: nucleoside 5-phosphate oxygen attacks P via SN2 associative, forming pentacoordinate TS, imidazole departs. Step 3: nucleoside attaches at terminal phosphate, chain reaction possible.

6 Nucleotide Biosynthesis

Purine nucleotides: Ribose–5 phosphate + ATP $\longrightarrow$ PRPP, pyrophosphate makes C1 electrophilic. Steps: displacement of pyrophosphate at C1 by ammonia + glutamine = β – D-5-phosphoribosylamine, inversion of configuration at C1 (displacement by PRPP amidotransferase)

10 steps: glycine addition (C4, C5 and N7) $\rightarrow$ glycinamide ribonucleotide, formylation to add a formyl group on C8 $\rightarrow$ α -N-formylglycinamide ribonucleotide, glutamine amination (adds N3), ring closure, carboxylation at C4 with CO_2 , aspartate addition (contributes N1), formylation (adds C2), second ring closure. Product = IMP

AMP branch:

- IMP activated by adenylosuccinate synthetase ($\text{IMP} + \text{GTP} \longrightarrow \text{adenylosuccinate}$)
GTP prevents competition of energy source between AMP and GMP;
- Aspartate donates N6
- Elimination of fumarate by adenylosuccinate lyase ($\text{AMP} + \text{fumarate} \longrightarrow \text{adenylosuccinate}$); product = AMP

GMP branch:

- Oxidation of IMP by IMP dehydrogenase;
- $\text{IMP} \longrightarrow \text{XMP}$ via NAD^+ ;
- Oxidation = electrophilic C2;
- Amination via GMP synthetase ($\text{XMP} + \text{ATP} + \text{Glu} \longrightarrow \text{GMP}$)

7 Biosynthesis of Pyrimidine Nucleotides

Carbamoyl phosphate = Glu + bicarbonate catalyzed by carbamoyl synthetase. Committed step = carbamoyl phosphate + ASP $\longrightarrow$ N – Carbamoyl aspartate (inhibited by UMP). Ring closed via cyclization + dehydration + dehydrogenation $\rightarrow$ orotic acid.

Deoxyribonucleotide formation:

1. Tyrosyl radical formation via **ribonucleotide reductase (class I)**. Reaction: O_2 oxidizes iron $\rightarrow$ tyrosyl radical. Triggers a chain of electron transfers;
2. Radical transfer to ribose: proton-coupled electron transfer moves radical from Tyr to Cys to ribose. Cys in active site = thiyl radical, attracts H from C3 of ribose, yielding a C3 radical.
3. Formation of the 2-hydroxyl leaving group: C3 radical causes electron rearrangement, hydroxyl activated, cys thiol protonates the 2-hydroxyl forming water, water leaves = 2-carbocation-like intermediate.
4. Reduction of 2-carbocation: Another cys donates hydrogen, converting 2-carbocation to deoxy position. Collapse of ribose radical to sugar.
5. Radical is transferred back from ribose to Cys to Tyr;
6. Thioredoxin reduces oxidized cys and reduced by thioredoxin reductase (electrons supplied by NADPH)

8 Ribonucleotide Reductases (RNR)

8.1 Class I

Aerobic, di-iron / tyrosyl-radical enzymes transforming ribonucleotides (NDP) into deoxyribonucleotides (dNDP). Mechanism = deoxyribonucleotide formation. Note: Allosteric regulation by ATP / dATP: ATP activates class I RNR, dATP inhibits by locking α_2 and β_2 into non-productive interface. ATP binding to two sites in cone domain reverses inhibition. Class I RNR = dioxygen-dependent di-iron chemistry.

8.2 Class II - adenosylcobalamin (AdoCbl) radical enzyme

Radical generator = AdoCbl coenzyme B_{12} , oxygen independent (anaerobic function possible), no tyrosyl radical, no di-iron center, radical generated directly from AdoCbl via homolytic cleavage of Co–C bond.

mechanism:

1. AdoCbl undergoes homolysis: $\text{AdoCbl} \longrightarrow \text{Co(II)} + 5\text{-deoxyadenosyl radical}$ catalyzed by enzyme;
2. Hydrogen abstraction: 5-dAdo abstracts H from C3, producing a C3-radical on ribose, 5-dAdo becomes 5-deoxyadenosine (5-dAdoH)
3. activation and departure of OH: radical rearrangement activates hydroxyl, cysteine protonates $\text{OH} \rightarrow \text{H}_2\text{O}$ forming carbocation intermediate
4. same as deoxyribonucleotide formation without PCET
5. Return of radical to AdoCbl: ribose radical abstracts H from 5-dAdoH, regenerating 5-Ado, recombines with Co(II) forming AdoCbl.

8.3 Class III

Definition:

- Anaerobic enzyme destroys the radical via oxygen;
- SAM activator generates glycy radical
- Glycyl radical enzyme (GRE) stores stable radical on a glycine
- Thiyl radical catalysis

Mechanism:

1. $\text{SAM} + 4\text{Fe-4S} \rightarrow$ homolytic cleavage, product = 5-deoxyadenosyl radical (5-dAdo). 5-dAdo abstracts H from a specific glycine $\rightarrow$ glycy radical, product = 5-deoxyadenosine.

2. Gly abstracts H from a cysteine, product = Cys-S (thiyl radical)
3. Cys-S removes H from C3 (produces C3 radical on ribose) → sugar activated for deoxygenation.
4. Radical rearrangement makes 2-OH a leaving group, Cys protonates OH to produce water leaving group, forming 2-carbocation like group.
5. second Cysteine donates H, carbocation reduced (2-CH), collapse of sugar radical.
6. Ribose radical attracts H from Cys, Cys-S quenched, Gly is restored
7. Thioredoxin / glutaredoxin regeneration

9 Thioredoxin Cycle

Purpose: Keeps RNR's catalytic cysteines in the reduced -SH state so that chemistry is possible.

Players:

- Thioredoxin (Trx)
- Thioredoxin reductase (TrxR)
- NADPH
- RNR active site Cys

Mechanism:

1. RNR oxidates cysteines, forming disulfide bond during NDP → dNDP. Cys must be reduced before next turnover.
2. Thioredoxin binding RNR: thioredoxin has Cys – Gly – Pro – Cys motif. Cys are in reduced dithiol form and thioredoxin docks on RNR's oxidized active site.
3. Thioredoxin reduces RNR: first thioredoxin cysteine attacks RNR's disulfide, product = mixed disulfide. Second cysteine resolves mixed disulfide, releasing RNR with two reduced cysteines. Product = oxidized thioredoxin.
4. Thioredoxin reductase reduces thioredoxin: using NADPH as electron donor yielding NADPH → NADP⁺. Electrons flow through FAD to active site. Result: cysteines produce thioredoxin disulfide. Thioredoxin then returns to reduced dithiol form.
5. Cycle repeats.

10 dNDP to dNTP phosphorylation

Step performed by nucleoside diphosphate kinase (NDPK) using ATP as phosphate donor.

Mechanism:

1. Phosphate transfer: not substrate-specific, NDPK works for all nucleoside diphosphates.
2. Ping-Pong mechanism with phospho-histidine intermediate: ATP donates a phosphate to a specific histidine residue on NDPK, product = phospho-His with dNDP binding and phospho-his transfers phosphate yielding dNTP. Histidine then returns to its non-phosphorylated form.
3. Reaction driven by ATP to ADP transfer with RNR ratio controls.

11 Properties of DNA

Bases: Adenine (A), cytosine (C), Uracil (U), Thymine (T), Guanine (G) In DNA, T replaces the U in RNA.

Ionisation: amino bases A, C, G become protonated on one of ring nitrogens. Tautomerism: switch between keto-enol and enamine-imine tautomers.

DNA transcription of A by T and of C by G is based on hydrogen bonding.

Watson-Crick base pairing: anti-orientation for both bases; bases must be planar to maximize hydrogen bonding. Patterns: A-T (DNA) / A-U (RNA) - A donates one hydrogen bond and accepts one, T or U accepts one and donates one. G-C: G donates two, accepts one, C donates one, accepts two.

11.1 Syn-Anti Conformation

Anti = H6 pyrimidine or H8 purine above sugar ring. Syn = O-2 (pyrimidine), N-3 (purine) above ring. C4-C5 orientation: synclinal or antiperiplanar rotamers.

12 DNA forms

B-DNA (cellular form): C2-endo pucker, anti orientation, right-handed helix, 3.4Å per base pair, 34 degree twist, 10.5 bp/turn with wide major groove and narrow minor groove. Stability due to hydration of C2-endo geometry, $\pi - \pi$ interactions dominate stability, backbone torsion angle = small, stable range.

A-DNA: C3 pucker, right-handed helix, 33 deg twist, shallow major groove, deep narrow minor groove.

Z-DNA: left-handed zigzag, alternating purine-pyrimidine sequences. C2 endo (pyrimidines), C3 endo (purines). Transient during transition and supercoiling.

Torsion angles: $\alpha = \text{O}_3' - \text{P}$, $\beta = \text{P} - \text{O}_5'$, $\gamma = \text{O}_5' - \text{C}_5'$, $\delta = \text{C}_5' - \text{C}_4'$, $\epsilon = \text{C}_4' - \text{C}_3'$, $\zeta = \text{C}_3' - \text{O}_3'$. δ determines sugar pucker.

Primary structure of DNA: $5' - \text{dpApGpCpTpG}$, A and G = purines, C, T, and U = pyrimidines.

Electrostatic interactions between bases: repulsive forces between the sugars and the bases and $\pi - \pi$ stacking interactions determined by electron density above and below the plane. Energy contributed by:

- Van der Waals forces
- Electrostatic interactions due to partial atomic charges
- Electrostatic interactions due to distribution of charges associated with $\pi - \pi$ electron density above and below the base plane
- Electrostatic interactions due to distribution of charge associated with pi-electron density and partial atomic charges.

Characteristic of structure dependent modulation: propeller twist (when bases rotate by 5 to 25 deg relative to their H-bonded partner along C8 of purine and C6 of pyrimidine).

Calladine's rules: Watson-Crick pairs impose fixed hydrogen-bond geometry. Each dinucleotide step has characteristic roll/tilt/twist/slide/shift/rise. Changing one parameter geometry forces compensatory changes in others. Distortions cannot break $\pi - \pi$ overlap or Watson-Crick geometry. Groove continuity must be preserved (minor/major grooves cannot jump between steps). Limitation = cross hydrogen bonding (N6 of adenine with opposite O4 of thymine) impossible to model.

DNA binding: bent DNA solution = bend of 26° in helix axis. Bending = at junctions with stiff dA-dT helix with a B-helix, result = continuous DNA curve.

Transition mismatches: purine with wrong pyrimidine and transversion mismatches: either two purines or two pyrimidines. To detect mismatches, only possible when given two strands (the original and its complement). Step 1: reverse strand 2 and compare the corresponding bases from both strands.

DNA curvature = sequence writhes + poly(dA) straightness.

Cruciform formation mechanism: S-pathway

1. Small local bubble formation: few base pairs at center of inverted repeat melt (low energy cost)
2. First arm nucleates: partial hairpin fold on one side (proto-cruciform)
3. Proto-cruciform region stabilization: melted region torsional stress reduced
4. Second arm nucleates: second partial hairpin fold and formation of full cruciform

C-pathway:

1. Large bubble formation: many base pairs melt across inverted repeat region (strong negative supercoiling)
2. Expansion of bubble to critical size: full exposition
3. Both arms nucleate and forms hairpins simultaneously
4. Bubble collapses into two hairpins lock into the four-way junction to form cruciform

Roles of metal ions in DNA structures: charge screening - Phosphates more compact than in B-DNA = repulsion. Metal ions bind near phosphates, lowers local electrostatic energy and allows tight packing without destabilization when reducing the repulsion forces. Direct coordination to bases and stabilizing unusual geometries.

A-tracts: narrow minor groove, high propeller twist, distinct B-form helix, cations in minor groove G-tracts: cations in major groove, B to A transition at higher ionic strength, wider deeper major groove. Generic DNA: cations in both grooves more evenly, B-form stabilization, no strong sequence-induced curvature.

B-Z junctions:

- Base extrusion: One Watson-Crick base pair at junction breaks; both bases out of helix.
- Stacking continues despite the break.
- Helix orientation switch from right handed (B-DNA) to left-handed (Z-DNA). Discontinuity resolved by junction.
- Extrusion relieves torsional strain caused by B-Z switch.

Extrusion = necessary because Z has alternating syn/anti conformation. B has anti only. Must break base pair for transition. Preference = A-T pairs.

B-Z transition requires:

- Negative supercoiling (by transcription, nucleosome unwrapping, topoisomerase activity), lowers energetic cost of Z-DNA forming.
- CG sequence repeats with AT next to CG site for extrusion.
- Z-DNA formation stabilized by protein.
- Every Z-segment is flanked by two B-Z junctions (one at each end)
- Z-DNA formation is reversible once stress dissipates.

DNA circularization is possible for eukaryotic DNA by holding DNA loops together by means of a protein scaffold. Vinograd observation: strained relaxed DNA circle = torsional strain helix by winding around itself to form a superhelix. DNA response to intercalation (inserting ethidium between DNA base pairs):

reduction of right-handed negatively supercoils until full relaxation and increase of left-handed supercoils (positively supercoiled).

Topological invariants:

- Linking number L_k - number of times one strand winds around the other.
- Twist T_w - helical turns of duplex.
- Writhe W_r - 3D coiling of the axis.
- Topological identity: $L_k = T_w + W_r$

Supercoiling is due to: fixed L_k unless breaking of a strand, underwinding or overwinding changes T_w , fixed $L_k \rightarrow$ change in T_w must be offset by a change in W_r , molecule adopts supercoils to minimize torsional strain.

Types of supercoiling:

- Negative supercoiling - underwound DNA, promotes strand separation, common in bacteria.
- Positive supercoiling - overwound DNA, stabilizes duplex, found ahead of replication forks.
- Plectonemic supercoils - intertwined. extended supercoils in bacterial plasmids.
- Toroidal supercoils - DNA around proteins, nucleosomes.

Consequences: replication stress: positive supercoils accumulate ahead of polymerases. Transcription stress - RNA polymerase generates positive supercoils ahead and negative behind. Local melting: negative supercoils promotes strand separation for initiation. Structural transitions - negative supercoiling can drive B to Z transitions in CG-rich regions.

Topoisomerases: Type I - cut one strand, change L_k by ± 1 . Type II - cut both strands, change L_k by ± 2 , manage replication stress.

Triple-helix DNA: two homo-pyrimidines + one homo-purine. Third-strand binding to an homo-purine-homo-pyrimidine duplex shows: a third homo-pyrimidine binds parallel to homo-purine by Hoogsteen hydrogen bonds (pyrimidine strands are anti-parallel). A third homo-purine binds anti-parallel to the original purine by reversed Hoogsteen hydrogen bonds. Bases in third strand = regular anti conformation in glycosylic bonding.

Triplex motifs: H-motif (pyrimidine third strand) - pyrimidine-rich, binds parallel to purine strand requires protonated cytosine ($C^+ \rightarrow$ pH dependent). Base triads: C^+-G-C and $T-A-T$. R-motif: third strand is purine-rich, binds antiparallel to the purine strand, stable at physiological pH, base triads: $A-A-T, G-G-C$.

Base triads: C^+G-C - protonated cytosine binds to the guanine of a G-C pair. $TA-T$ - thymine binds to adenine of an A-T pair. $GG-C$ - guanine binds via reverse Hoogsteen geometry.

Structural features: major groove of B-DNA contains the third strand. Triplex formation compresses duplex helix. Metal ions stabilize triplexes.

13 RNA Structure

Primary RNA structure: linear sequence of ribonucleotides connected by 3'→5' phosphodiester bonds with a 5 → 3 chain direction. 3-OH of one ribose attacks phosphorus of a nucleotide triphosphate forming a phosphodiester bond between 3-O of nucleotide and 5-phosphate of the next. Backbone = 3-O-P(=O)(O)-O-5 in the 5-3 direction. Full primary structure description: 5-pppG-A-U-G-C-U-G-A-OH-3, ie ribose - C1 base - phosphate repeat.

2-OH cleavage mechanism: OH⁻ deprotonates the 2-OH, 2-O⁻ attacks phosphorus, backbone snaps accelerated by Mg²⁺.

13.1 Secondary RNA Structure

A-RNA: standard double-stranded helix defined by C3-endo sugar pucker, right-handed helix, 11 bp per turn, deep narrow major groove and wide shallow minor groove, strong base tilt relative to axis.

A'-RNA: right-handed, C3-endo, longer helical pitch, reduced tilt, altered groove widths.

RNA interference: appears when long double-stranded RNA appears in cytoplasm, enzyme Dicer recognizes dsRNA and cleaves it into siRNA duplexes, siRNA duplexes are loaded into RNA-induced silencing complex (RISC) where passenger strand is degraded only guide strand stays bound to Argonaute. RISC uses guide strand to find complementary mRNA, silencing via 1. mRNA cleavage (Argonaute cuts mRNA at a specific site to degrade it) or 2. translational repression (RISC binds out without cutting, blocking translation).

Modified base RNA:

1. Pseudourine (Ψ - structure: C5 glycosylic linkage instead of N1. Extra H-bond donor N1-H available. Increased stacking improves base planarity. Helices stabilized. Change due to Ψ C-glycosidic bond in electronic distribution increase base stacking and backbone rigidity.
2. Methylated bases (m^5C , m^6A , m^1A , m^7G : m^6A = N6-methyladenosine; destabilizes A-form locally. m^5C = 5-methylcytidine; increases stacking. m^1A : disrupts Watson-Crick pairing. m^7G : cap structure, positive charge.
3. Inosine (I): formed by deamination of adenosine. Wobble pairing - I pairs with U, C, A; expands decoding.
4. Dihydrouridine (D): D formation reduces C5=C6 double bond. Loss of aromaticity weakens stacking. Increases backbone flexibility.

RNA-DNA duplexes - reverse transcriptase mechanism:

1. Primer binding: In the site, host tRNA is prepackaged in the virion and anneals to the site on viral RNA. Reverse transcriptase uses the 3-end of this tRNA as the primer for DNA synthesis.

2. Minus-strand DNA synthesis: RT synthesizes DNA from the tRNA primer towards the 5-end of the viral RNA, producing -sssDNA (minus-strand strong-stop DNA).
3. RNase H degrades RNA strands of the RNA:DNA hybrid behind the polymerase, clears way for DNA transfer.
4. First template switch: newly synthesized -sssDNA contains the R region copied from the 5-end of viral RNA. Anneals to the R region at the 3-end of the same RNA genome (first strand transfer)
5. Minus strand DNA synthesis continuation completes: RT continues synthesis of minus-strand DNA all the way to the 5-end of RNA template and RNase H continues RNA degradation.
6. Polypurine tract (PPT) resists RNase H cleavage, RT uses PPT to start plus-strand DNA synthesis.
7. Second template switch: plus-strand DNA contains U3 region, annealing to the U3 region at the 5-end of -sssDNA.
8. product = linear dsDNA with LRTs at both ends.

RNA-DNA hybrids adopt A-form because RNA's 2-OH forces C3-endo, DNA follows. Compressed major groove less accessible, expanded minor groove more accessible. Thermodynamic stability: RNA-RNA ζ RNA-DNA ζ DNA-DNA.

Bulges: one or more unpaired nucleotides on one strand. Internal loops: unpaired nucleotides on both strands. Consequences: bulges introduce local kinks, minor groove widens locally, unpaired bases reduce stacking, proteins bind bulges and loops.

Hairpins: inverted repeats fold back to form a stem and loop of A-form stem geometry. Consequences: loop destabilizes stacking, stem stabilizes, tetraloops are formed.

RNA triplexes: third strand binds in major groove, existence of non-Watson-Crick hydrogen bonds on polypurine / polypyrimidine tracts. Consequences: A-form major groove deep and narrow Hoogsteen pairing accessible, stabilized by magnesium ion, requires compatible Hoogsteen donor-acceptor pairs.

References

- [1] BLACKBURN, Michael G., GAIT, Michael J., LOACKES, David, WILLIAMS, David M.; *Nucleic Acids in Chemistry and Biology*; 3rd ed.; RSC Publishing; 2006; pp.13-136, 209-379.